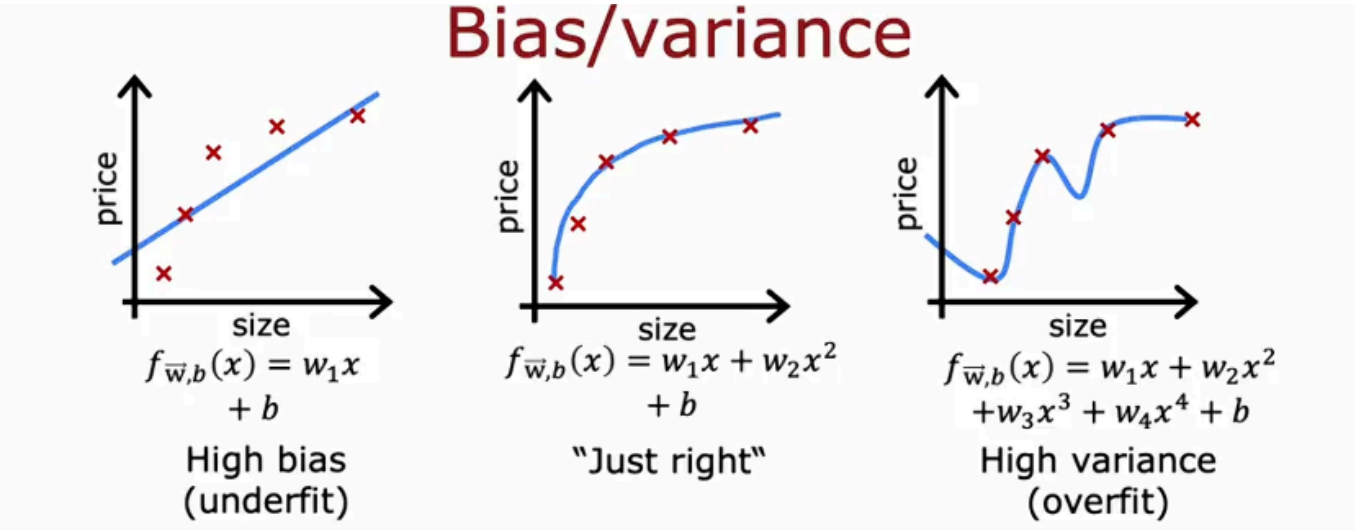
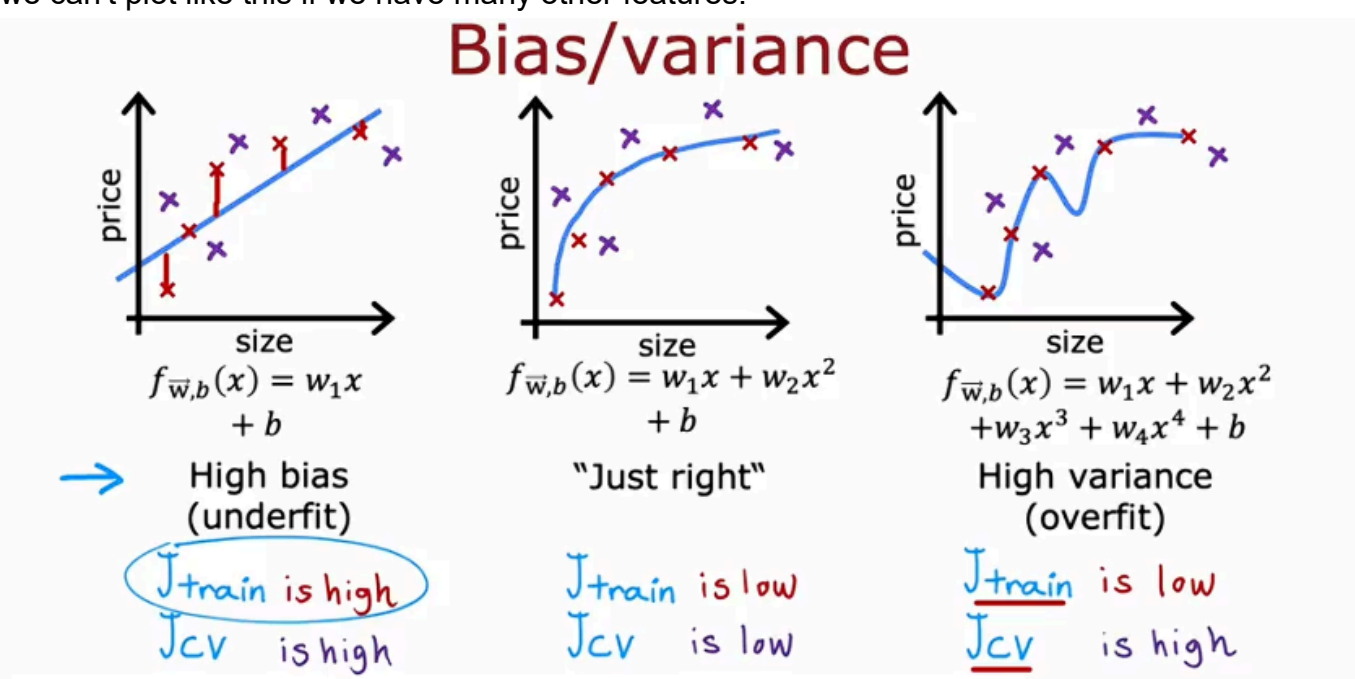
Day 21. Diagnosing bias and variance

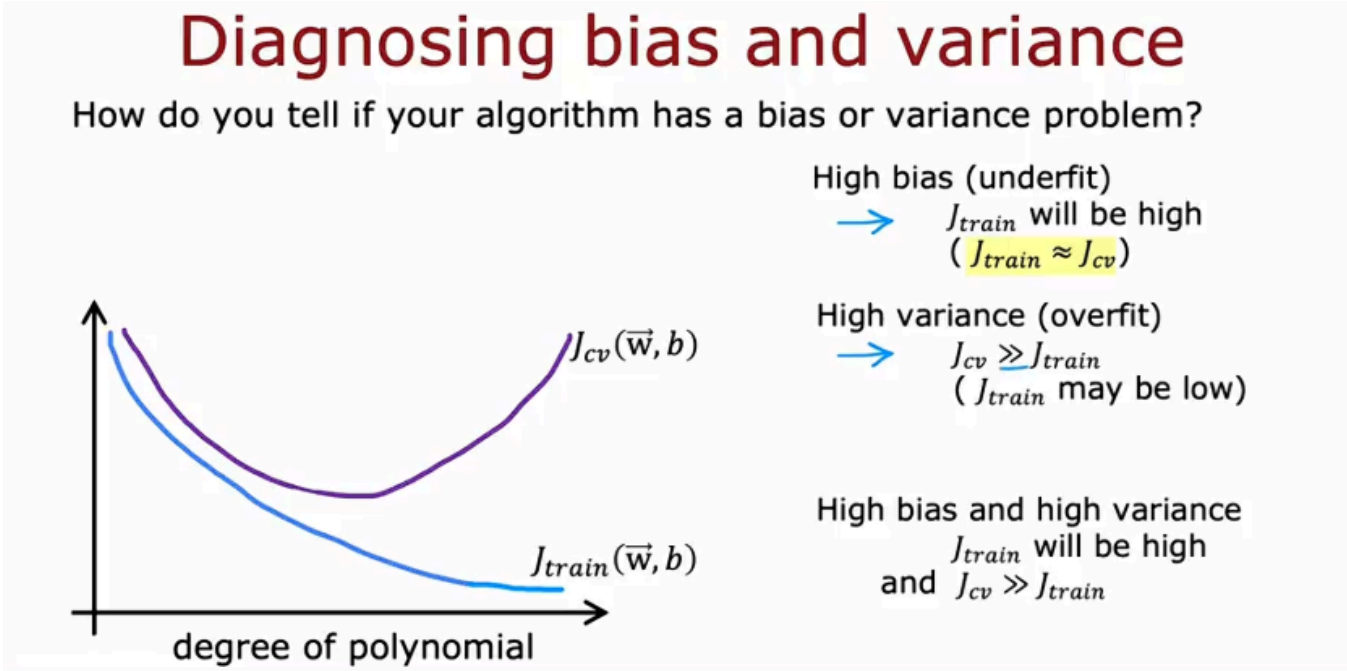
Bias and variance

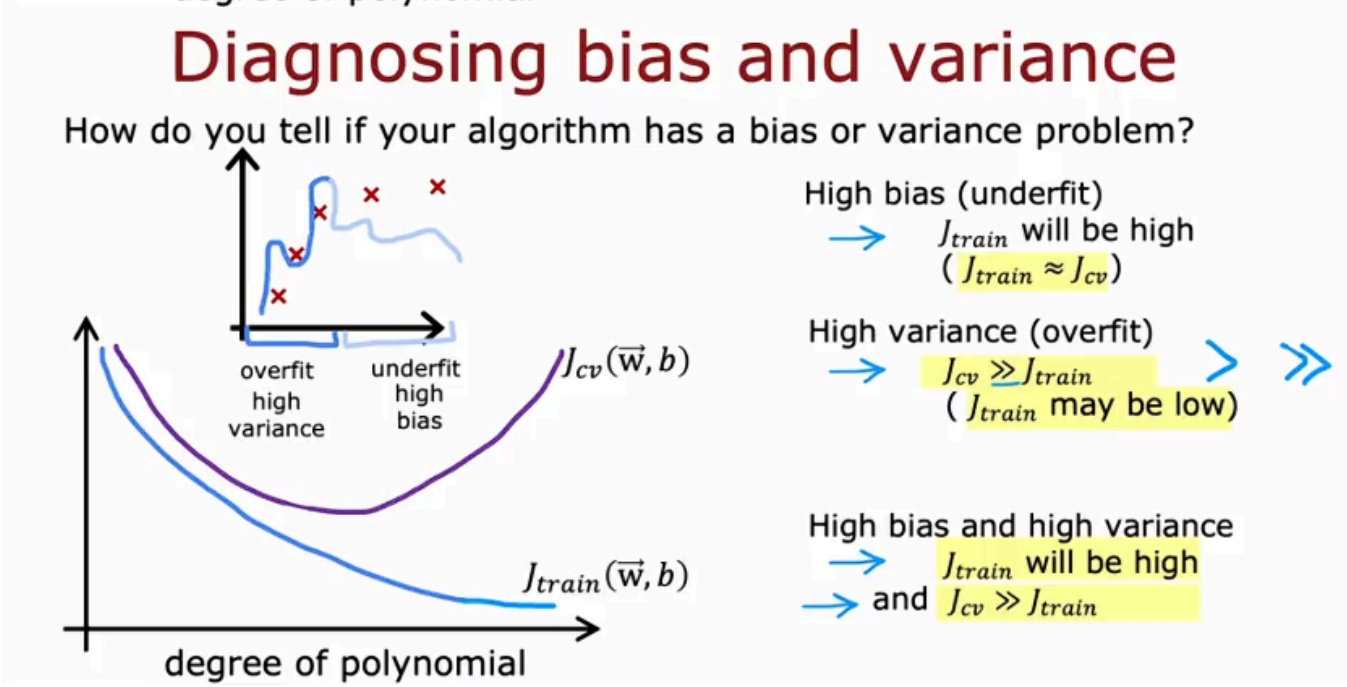
the typical workflow of developing a machine learning system is that you have an idea and you train the model, and you almost always find that it doesn't work well. Key to decide the process of building machine learning system is how to decide what to do next, and looking at bias and variance of a learning algorithm gives a good guidance on what to try next.



we can't plot like this if we have many other features.







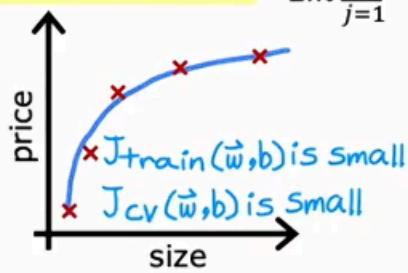
Regularization and bias/variance

Linear regression with regularization

Model: $f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

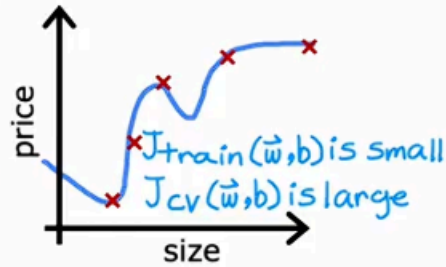
$$J(\vec{w},b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

Large λ
High bias (underfit)
$\lambda = 10{,}000$ $w_1 \approx 0, w_2 \approx 0$
$f_{\vec{w},b}(\vec{x}) \approx b$



Intermediate λ

λ



Small λ
High variance (overfit)
$\lambda = 0$

Choosing the regularization parameter λ

Model: $f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

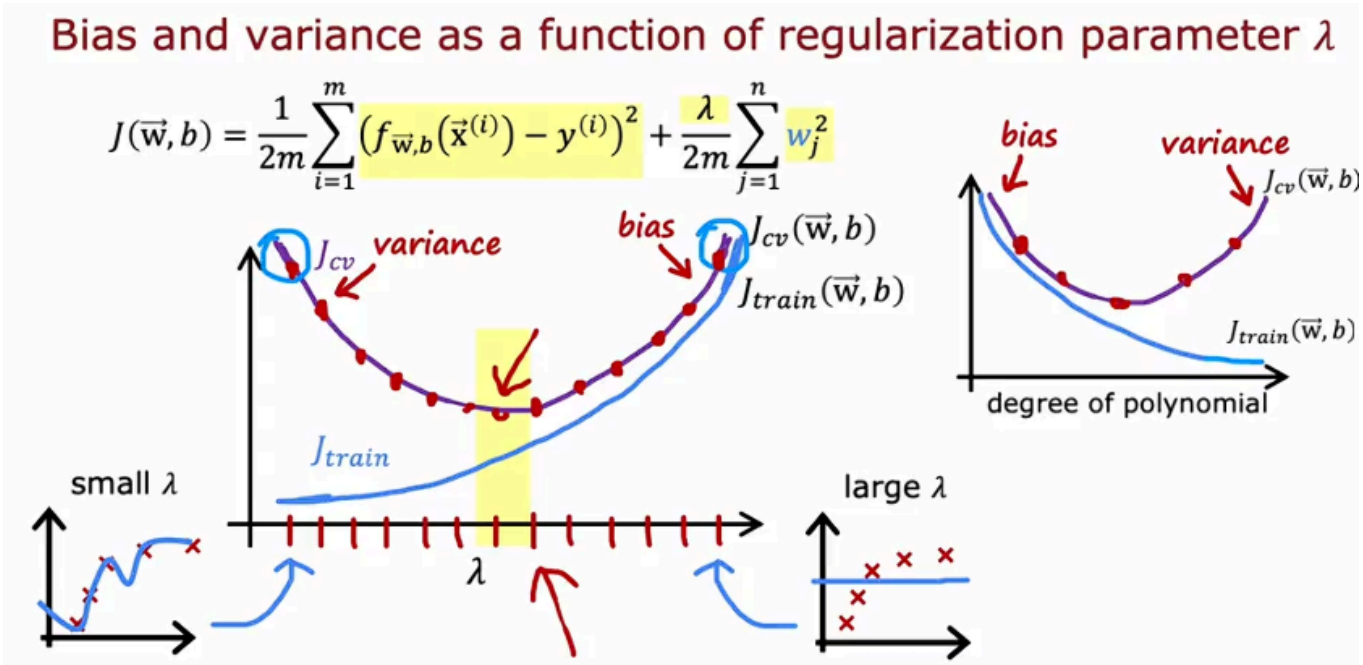
1. Try $\lambda = 0$ → $\min_{\vec{w},b} J(\vec{w},b)$ → $w^{<1>}, b^{<1>}$ → $J_{cv}(w^{<1>}, b^{<1>})$
2. Try $\lambda = 0.01$ → $w^{<2>}, b^{<2>}$ → $J_{cv}(w^{<2>}, b^{<2>})$
3. Try $\lambda = 0.02$ → $J_{cv}(w^{<3>}, b^{<3>})$
4. Try $\lambda = 0.04$
5. Try $\lambda = 0.08$ → $J_{cv}(w^{<5>}, b^{<5>})$

⋮

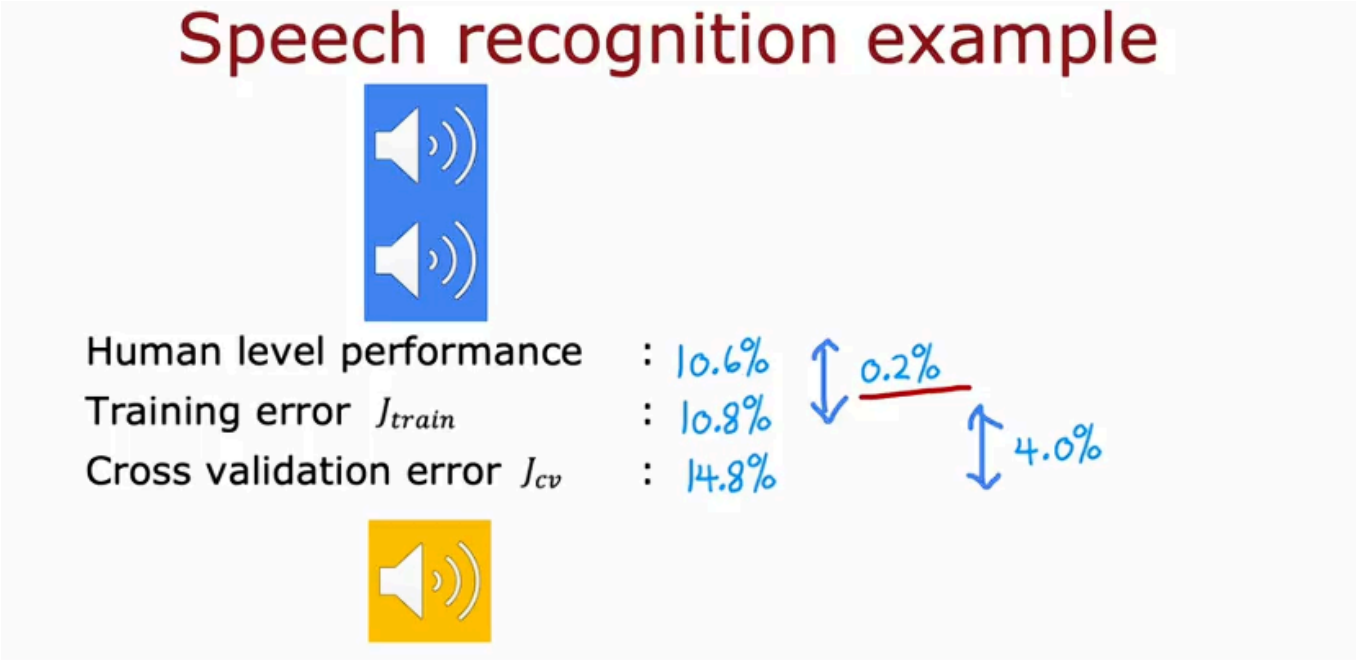
12. Try $\lambda \approx 10$ → $w^{<12>}, b^{<12>}$ → $J_{cv}(w^{<12>}, b^{<12>})$

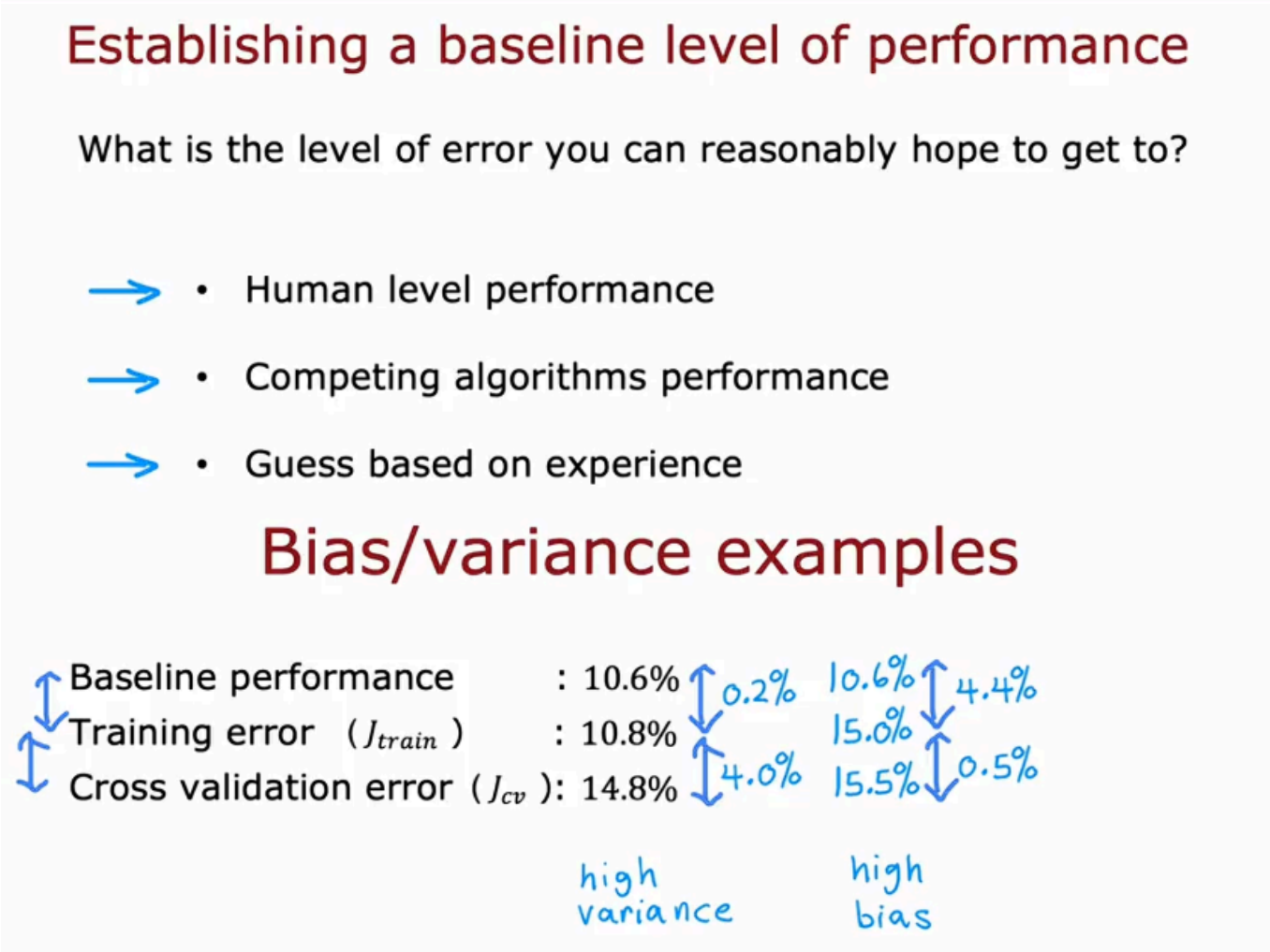
Pick $w^{<5>}, b^{<5>}$

Report test error: $J_{test}(w^{<5>}, b^{<5>})$



Establishing a baseline levle of performance





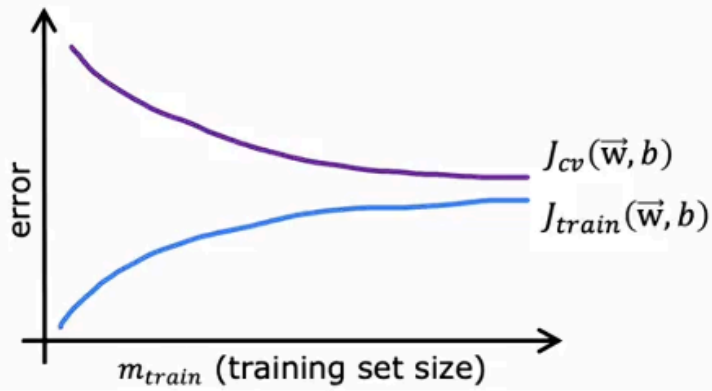
Learning curves

Learning curves are a way to help understand how your learning algorithm is doing as a function of the amount of experience it has, whereby experience, I mean, for example, the number of training examples it has.

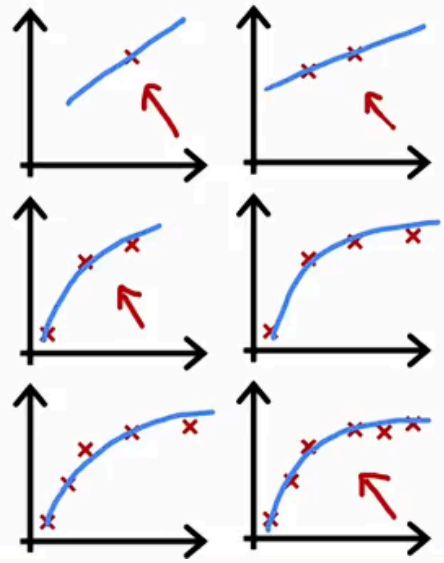
Learning curves

J_{train} = training error

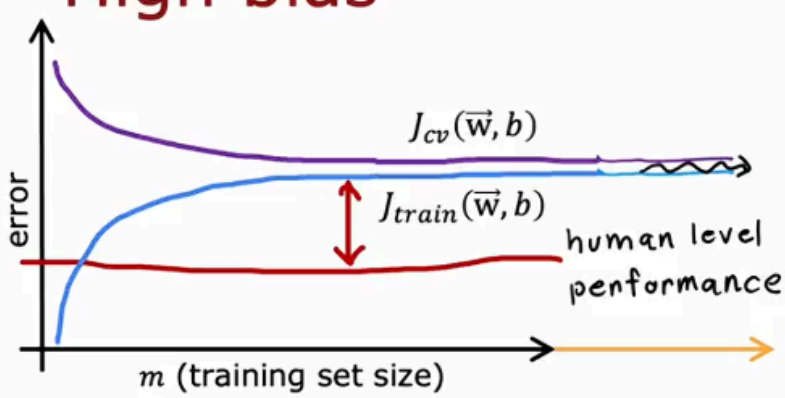
J_{cv} = cross validation error



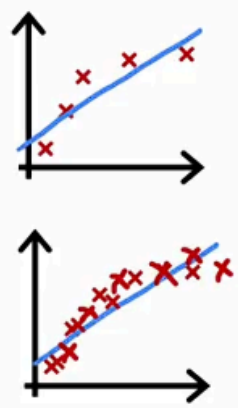
$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$



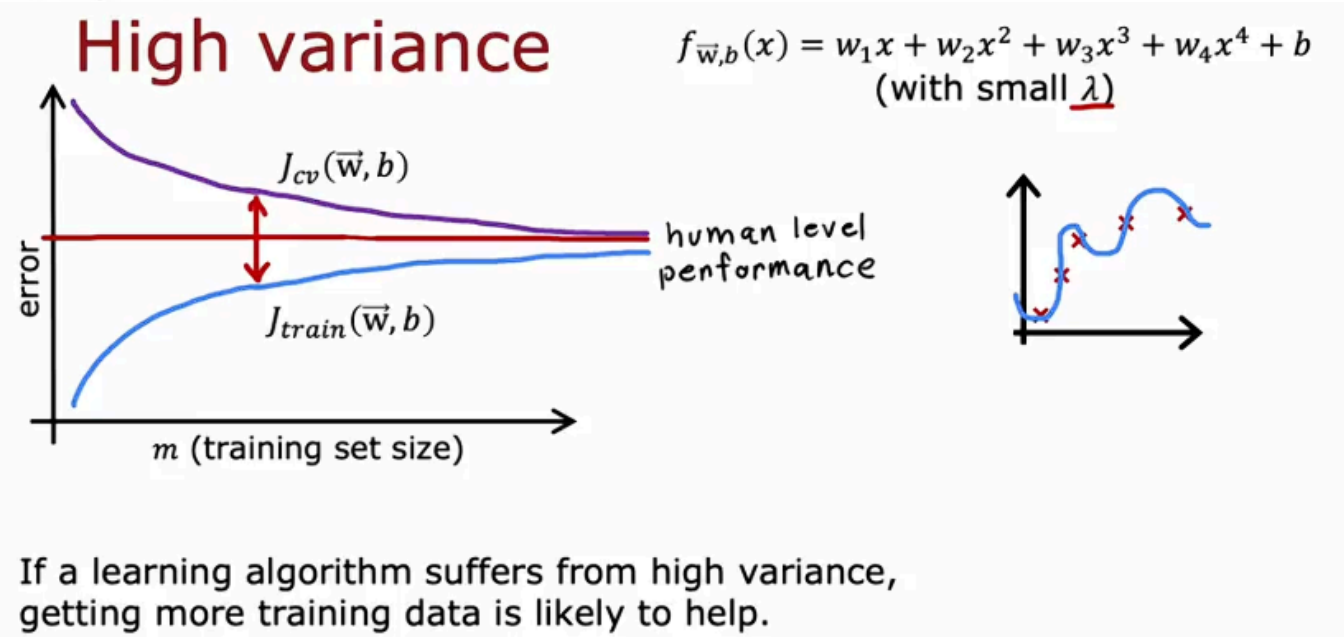
High bias



$f_{\vec{w},b}(x) = w_1 x + b$



if a learning algorithm suffers from high bias, getting more training data will not (by itself) help much.



If a learning algorithm suffers from high variance, getting more training data is likely to help.

Deciding what to try next revisited

recall these methods we have metioned at the beginning this week.

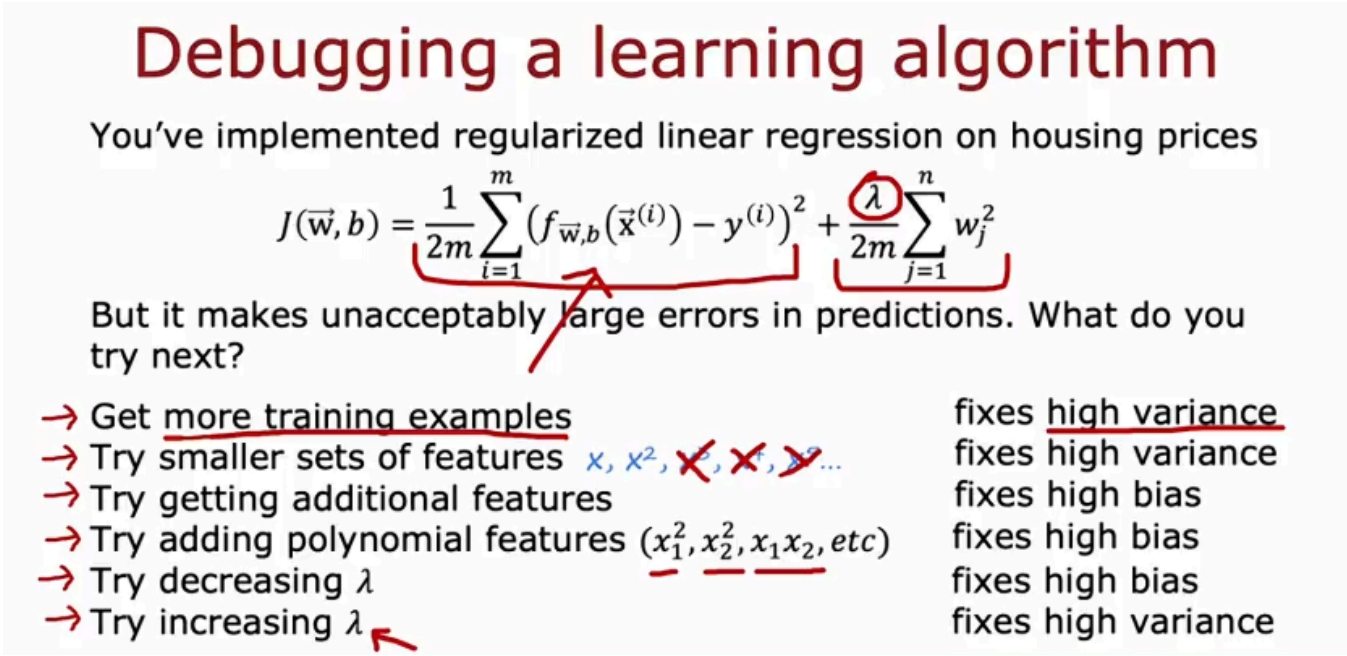
Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

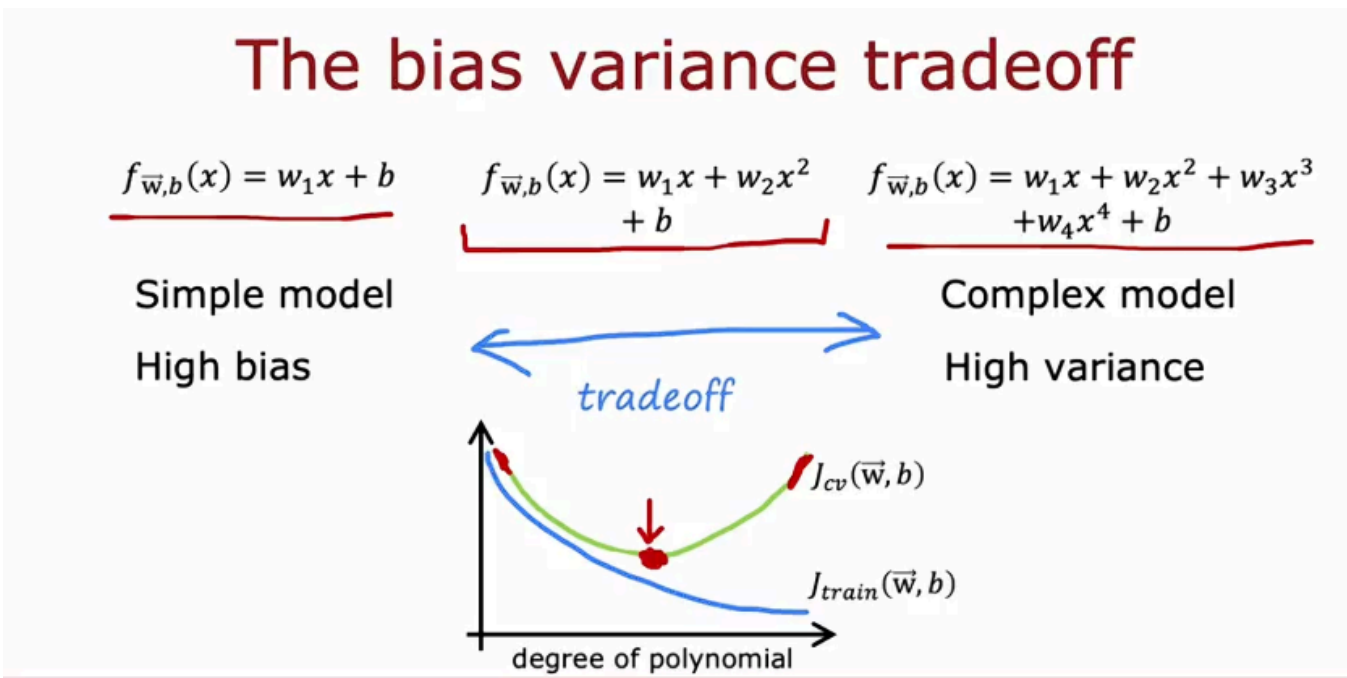
$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

But it makes unacceptably large errors in predictions. What do you try next?

- Get more training examples
- Try smaller sets of features
- Try getting additional features
- Try adding polynomial features $(x_1^2, x_2^2, x_1x_2, etc)$
- Try decreasing λ
- Try increasing λ



Bias/variance in neural networks



we will leave this part to next note.